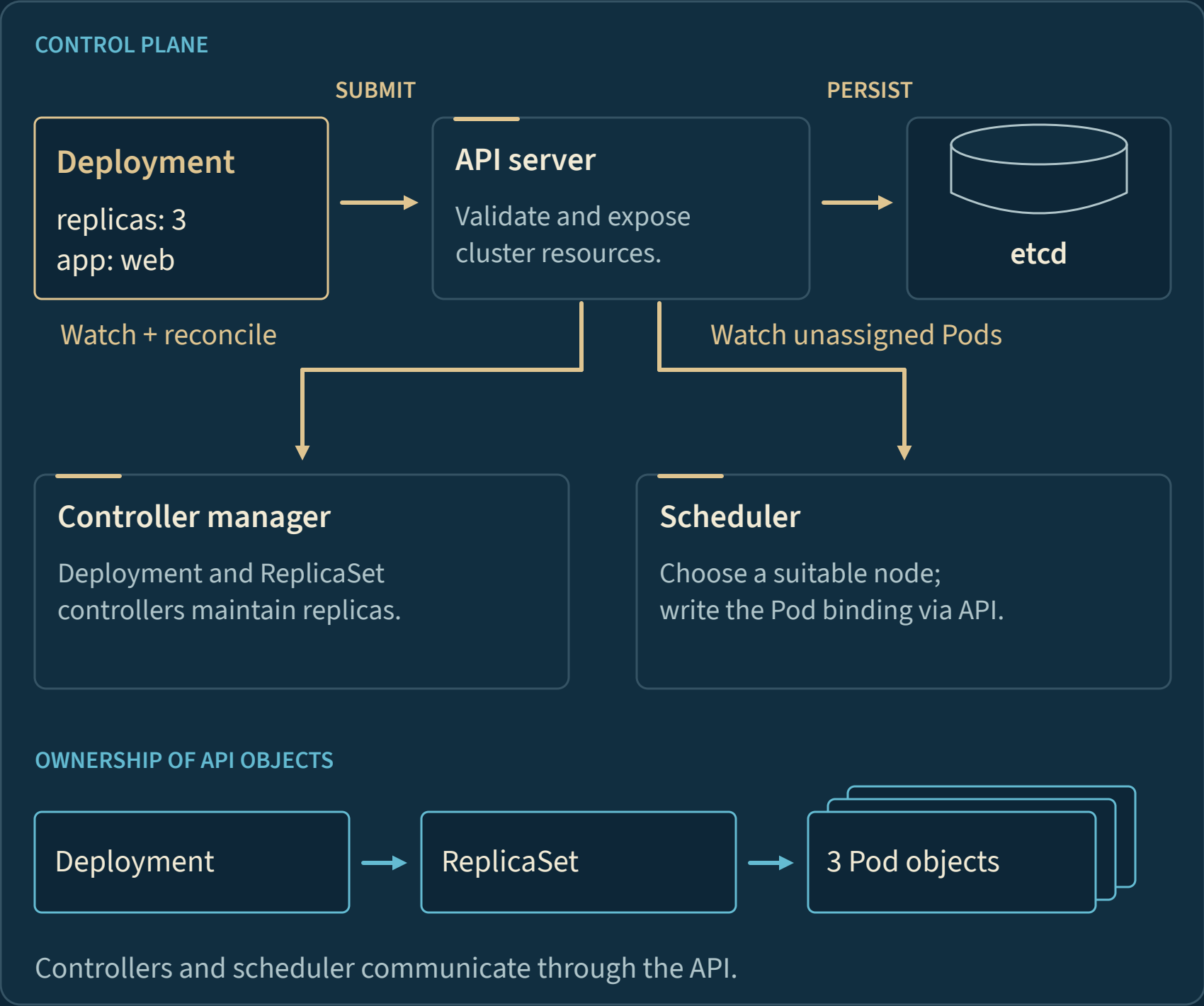


# Kubernetes: from declared replicas to live traffic

Four connected views of the same workload: control, execution, networking and replacement. Example: a three-replica web Deployment.

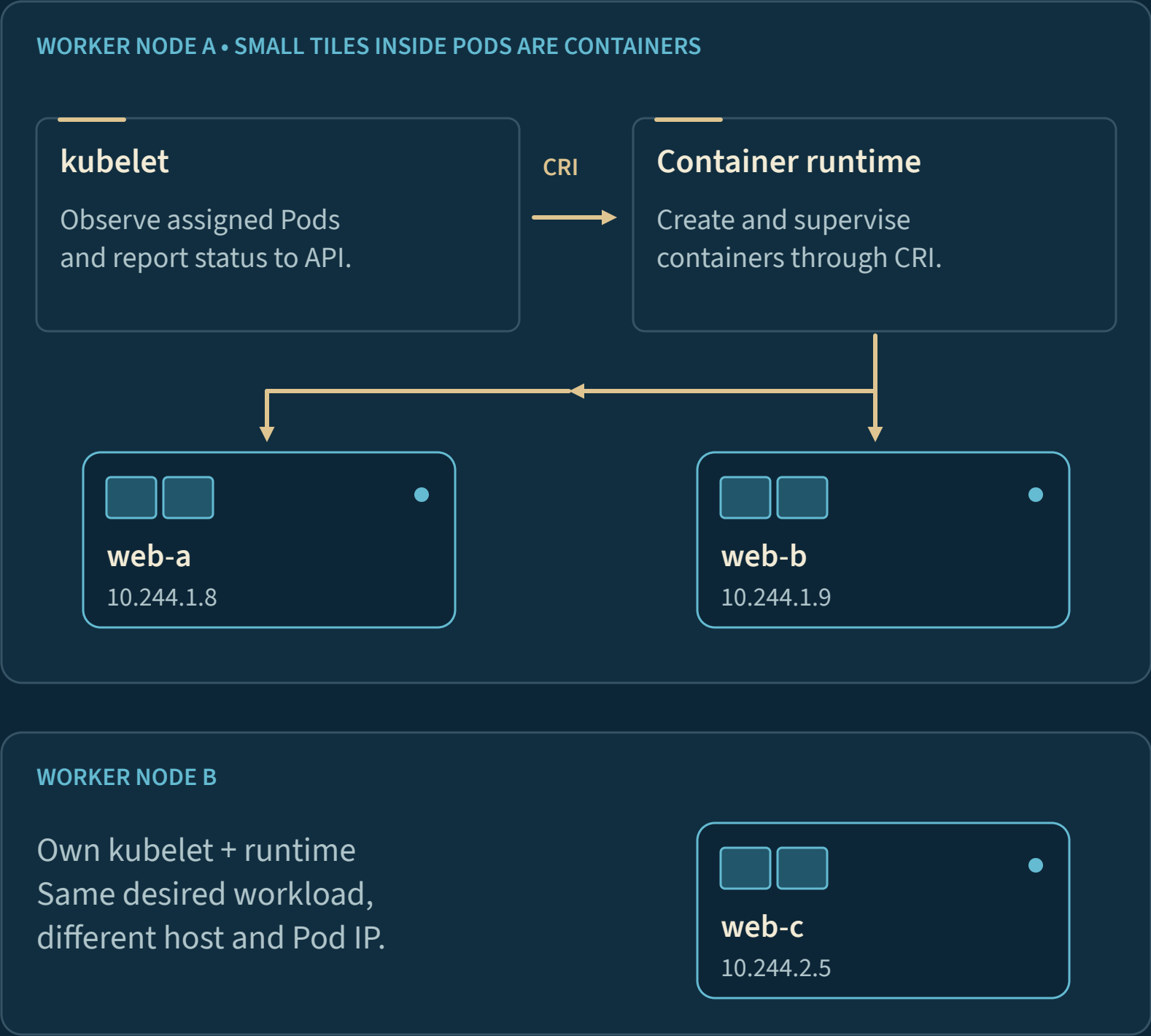
## 01 Declare intent; reconcile state

A Deployment asks for three replicas. Controllers keep that intent alive.



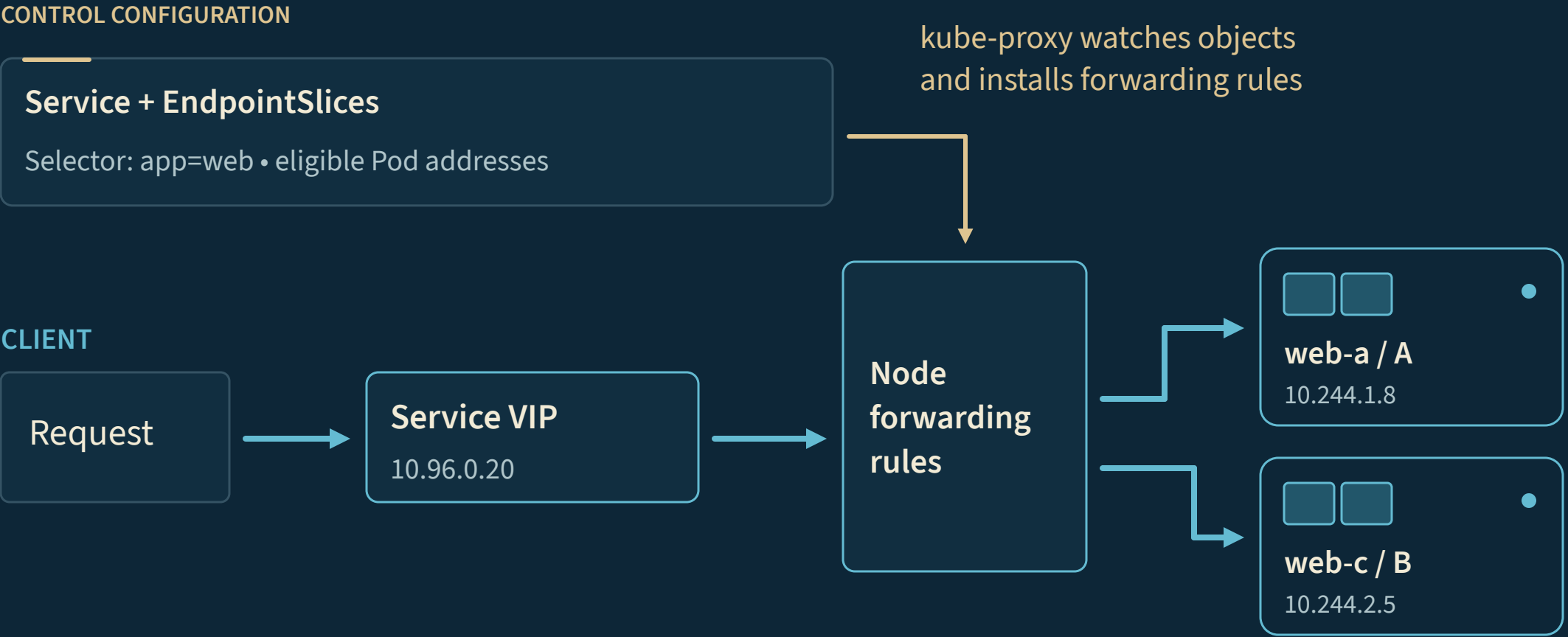
## 02 Run the assigned Pods

Kubelets observe bindings; runtimes create containers on each node.



## 03 A stable Service fronts changing endpoints

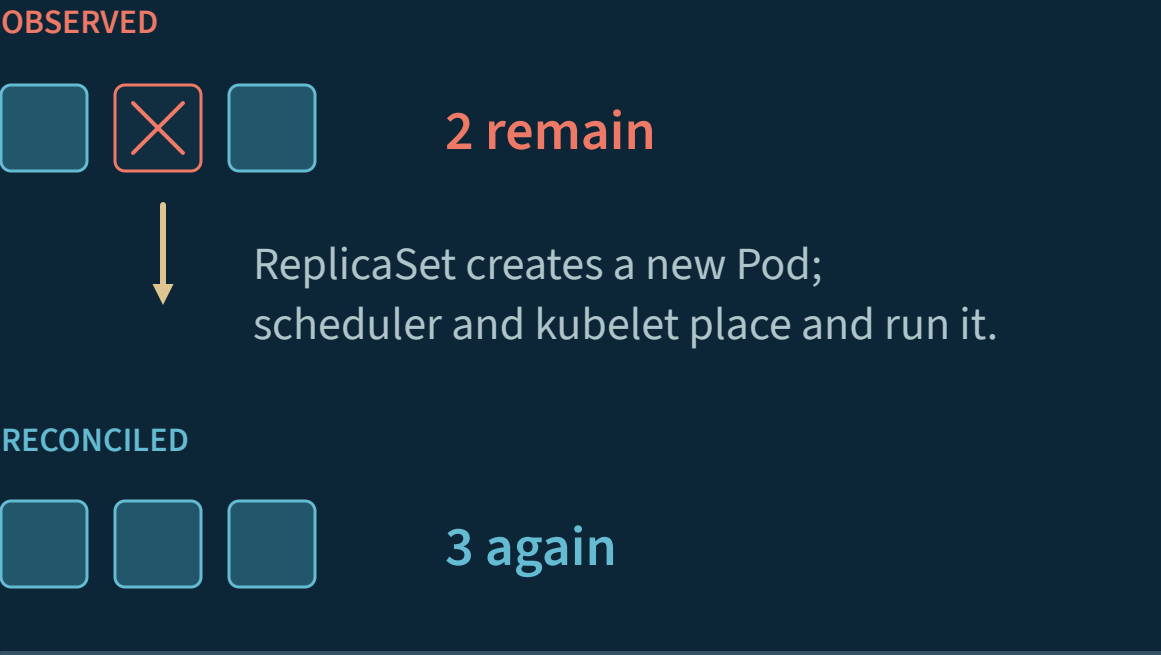
Application traffic follows the data plane. It does not traverse the API server.



One eligible endpoint per connection • two possible targets shown

## 04 Replace a lost Pod

The replica target stays at three. Pod identity and addresses can change.



EndpointSlices update as endpoints change.  
The Service address remains stable.